VENDORGATE

Vendor Onboarding and Purchase Order Portal

UML & SYSTEM DESIGN

Case Study No. 88

1. Introduction

This document defines the UML and system design view of VendorGate. It represents the major actors, system entities, interactions, workflows, and Purchase Order lifecycle in a clear visual form.

The actual diagrams will be created separately in Draw.io and inserted into the marked placeholders.

2. System Overview

VendorGate is intended to support the vendor registration and Purchase Order approval processes of the pharmaceutical distributor. The case study identifies Vendors, Procurement, Finance, and Audit as key stakeholders.

The UML design focuses on two major functional areas:

- Vendor Registration — supporting the movement from the existing email-based registration process to an online registration workflow.
- Purchase Order Management & Approval — supporting PO creation and approval according to the defined approval limits.

Approval levels defined in the case study:

Approval Level	Purchase Order Value
Level 1	Up to ₹50,000
Level 2	Above ₹50,000 up to ₹2,00,000
Level 3	Above ₹2,00,000 up to ₹10,00,000
Level 4	Above ₹10,00,000

3. UML Design Overview

- Use Case Diagram — actors and major interactions.
- Class Diagram — main entities/classes and relationships.
- Sequence Diagram — Raise and Approve a Purchase Order.
- Activity Diagram — workflow and decision points.
- Purchase Order State Diagram — Purchase Order lifecycle.

4. Use Case Diagram

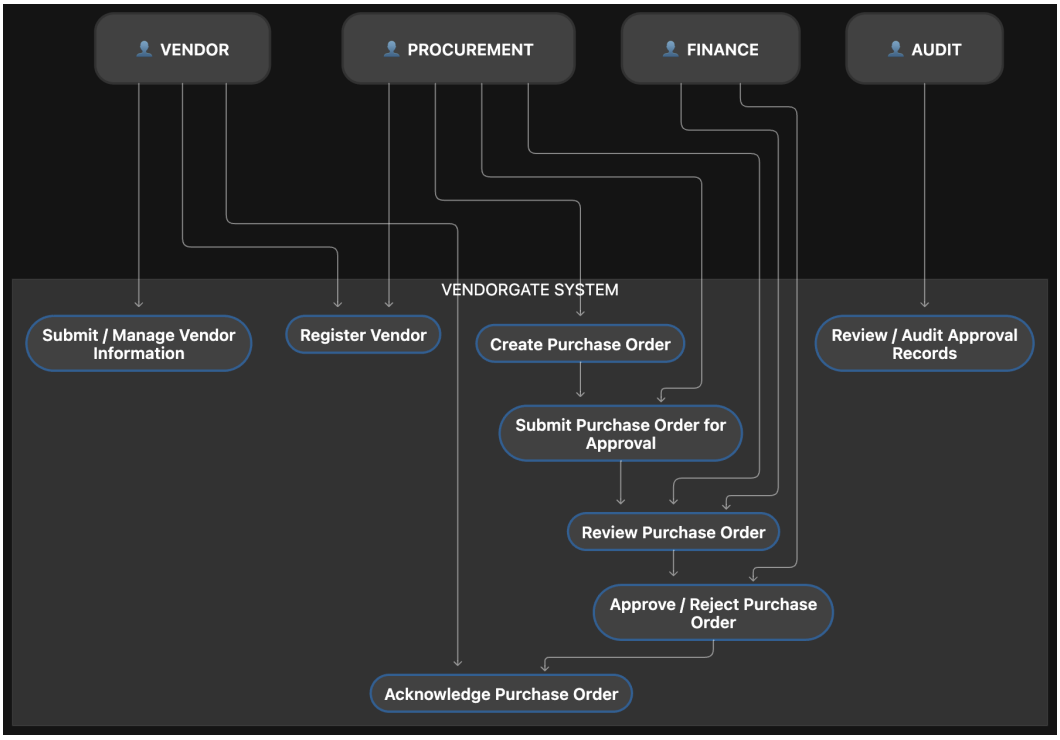
The Use Case Diagram gives a high-level view of how the identified stakeholders interact with VendorGate. It should focus on major business interactions rather than implementation details.

Primary actors:

- Vendor — vendor registration and acknowledgement activities.
- Procurement — vendor and Purchase Order processes.
- Finance — Purchase Order approval activities.
- Audit — review and monitoring of approval-related activities.

Major use cases to consider:

- Register Vendor
- Submit / Manage Vendor Information
- Create Purchase Order
- Submit Purchase Order for Approval
- Review Purchase Order
- Approve / Reject Purchase Order
- Acknowledge Purchase Order
- Review / Audit Approval Records

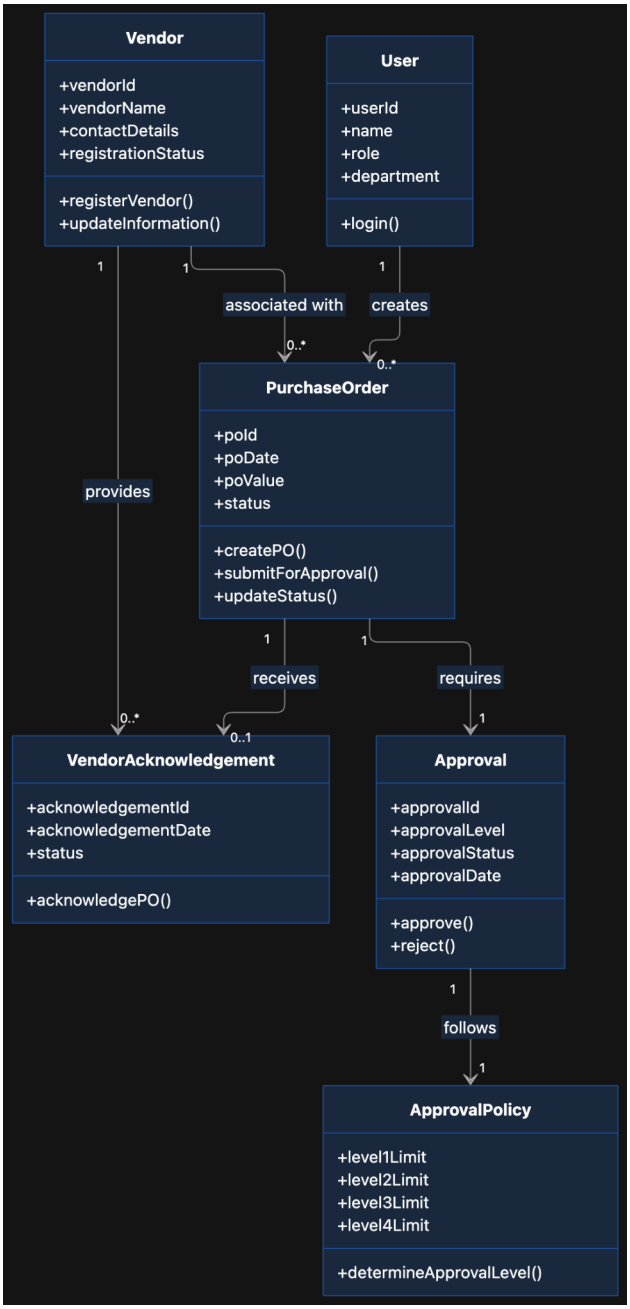


5. Class Diagram

The Class Diagram represents the main logical entities required to support VendorGate. Show important attributes and operations only where useful for explaining the design.

- Vendor — vendor information and registration details.
- User / Stakeholder — users involved in system activities.
- PurchaseOrder — Purchase Order and its value/status.
- Approval — approval action and approval level.
- ApprovalPolicy — defined Purchase Order approval limits.
- VendorAcknowledgement — vendor acknowledgement of a Purchase Order.

Relationships should make the responsibilities of Vendor Registration and PO Approval clear without tightly coupling the two functional modules.



6. Sequence Diagram — Raise and Approve a Purchase Order

This required sequence diagram represents the process of raising and approving a Purchase Order and the order of interactions between the relevant actor and system components.

1. Procurement initiates creation of a Purchase Order.

2. The system captures the Purchase Order details and value.

3. The system determines the applicable approval level using the approval-limit policy.

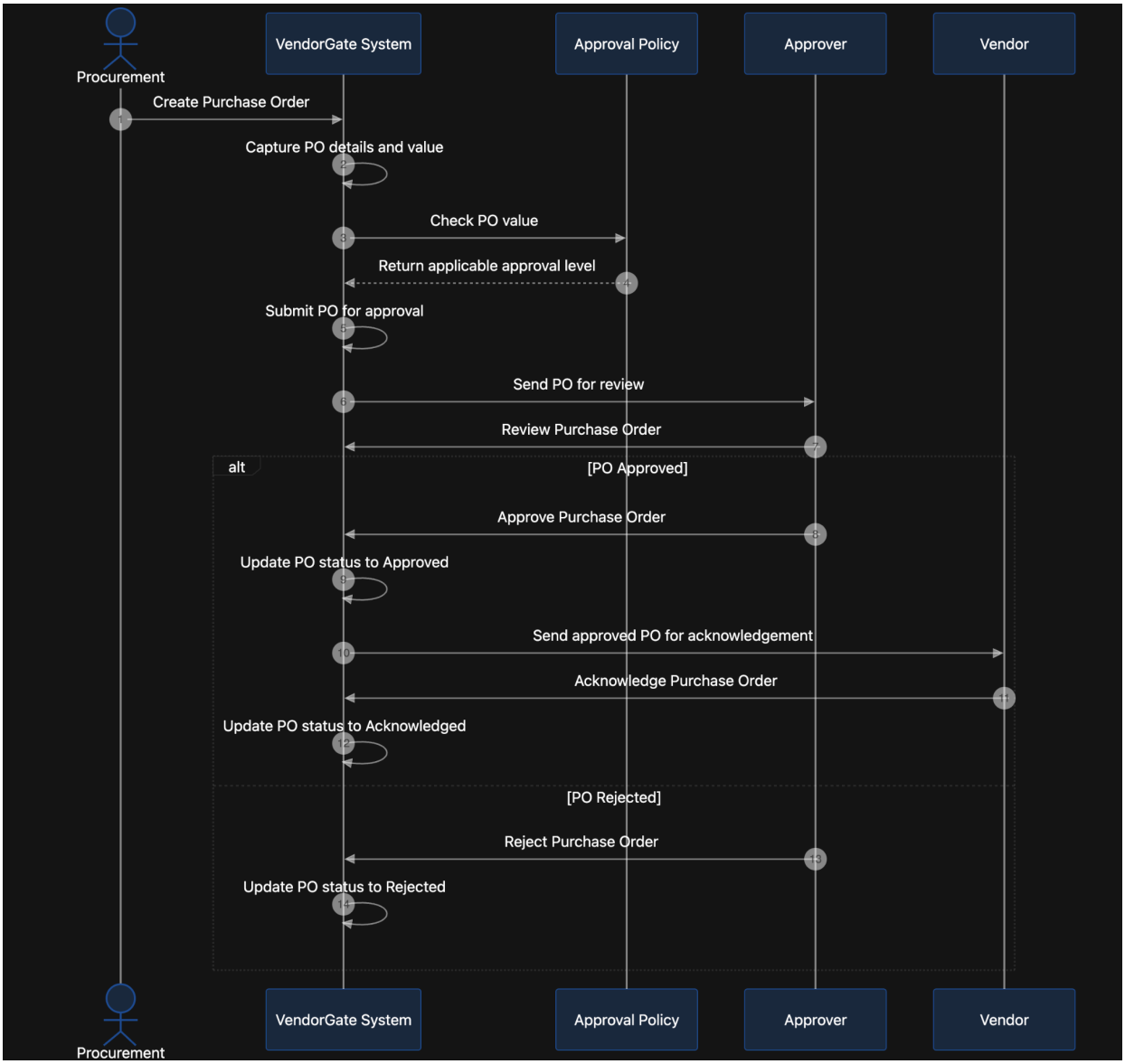
4. The Purchase Order is submitted for approval.

5. The appropriate approver reviews the Purchase Order.

6. The approver approves or rejects the Purchase Order.

7. The system records the decision and updates the Purchase Order status.

8. Where applicable, the Vendor receives the Purchase Order for acknowledgement.

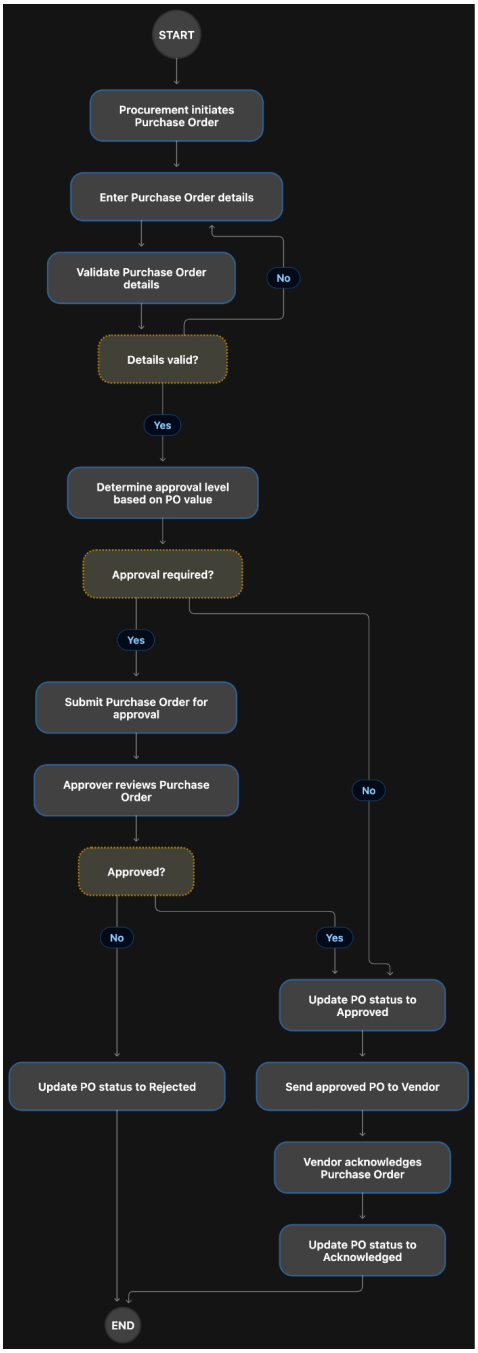


7. Activity Diagram / Workflow Diagram

The Activity Diagram should show the selected VendorGate workflow from initiation to completion, including important decisions and alternative outcomes.

- Process initiation
- Data entry/submission
- Validation
- Approval-level decision
- Approval or rejection path
- Vendor acknowledgement where applicable

- Process completion

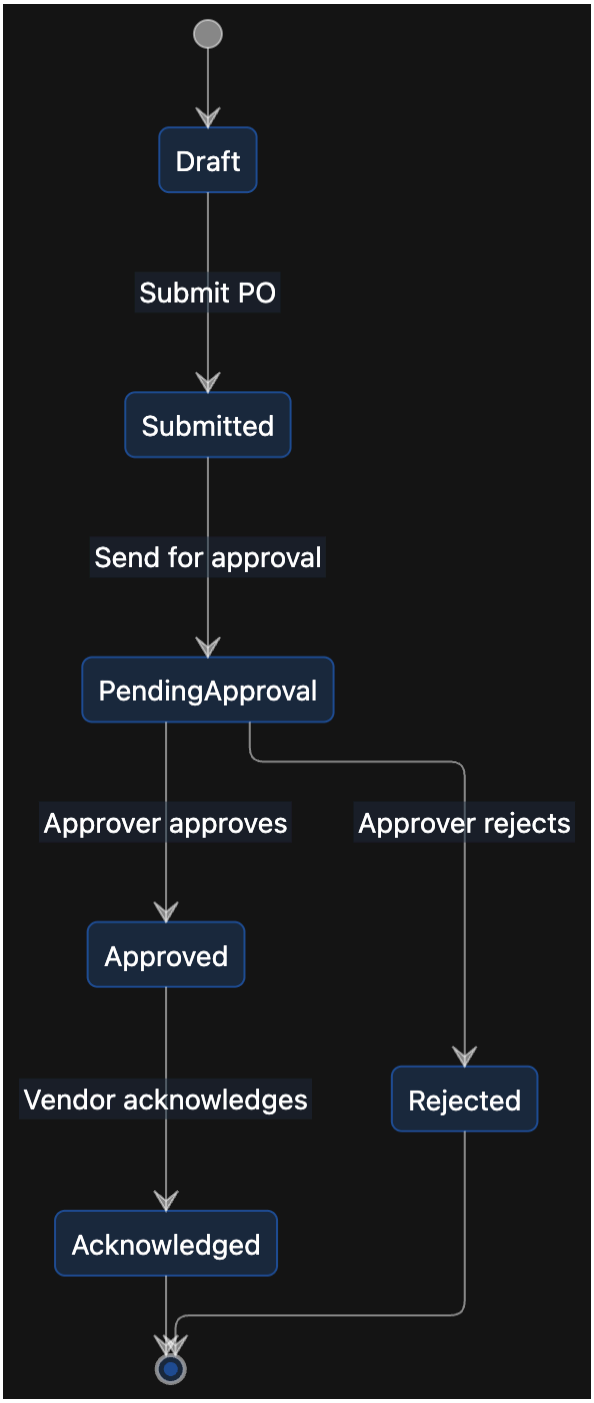


8. Purchase Order State Diagram

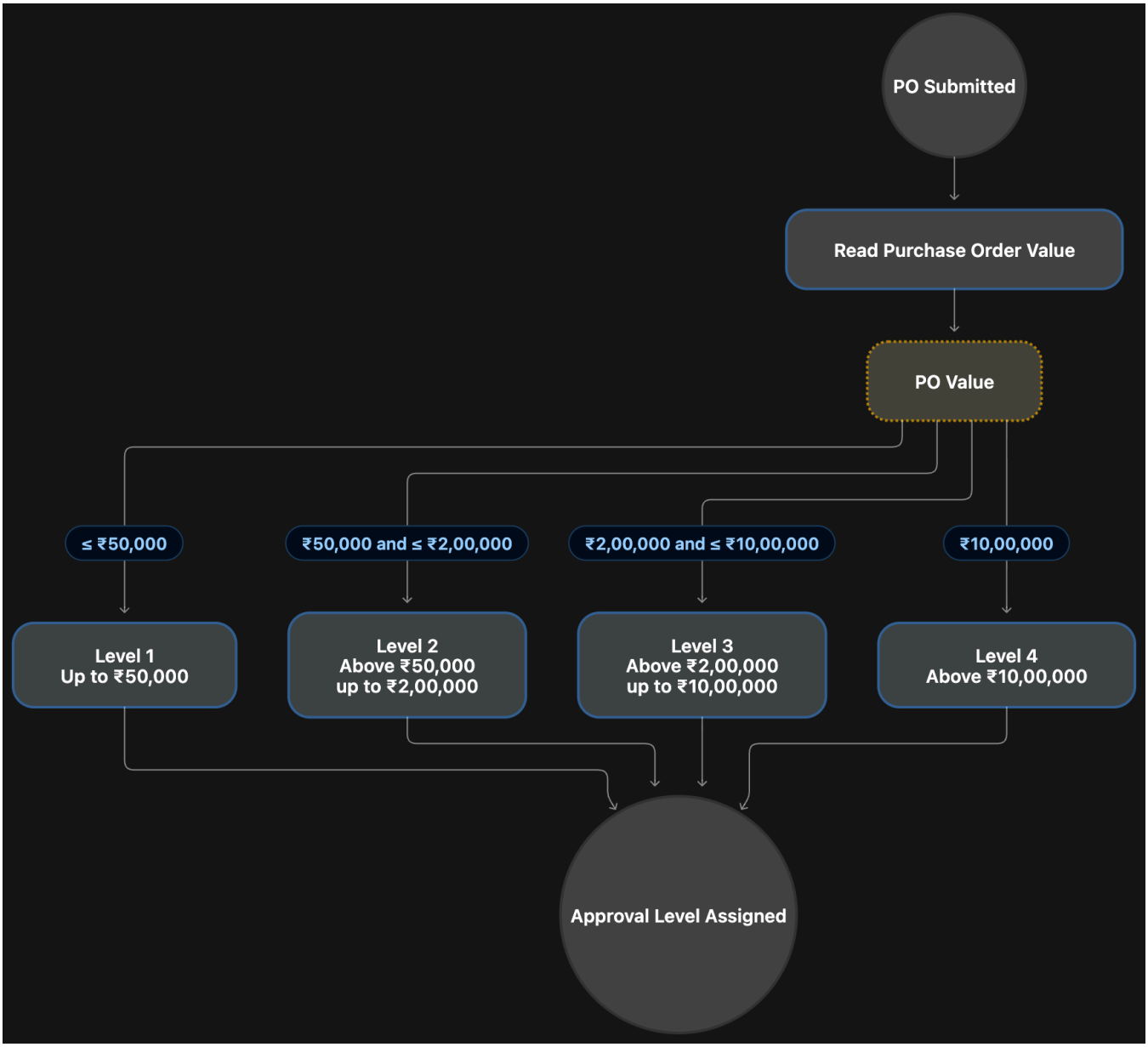
The Purchase Order State Diagram represents how a Purchase Order changes state during its lifecycle. The final state names should remain consistent with the prototype workflow.

- Draft — Purchase Order is being prepared.
- Submitted — Purchase Order has been submitted for approval.
- Pending Approval — Purchase Order is waiting for the required approval.
- Approved — Purchase Order has received the required approval.
- Rejected — Purchase Order has been rejected.

- Acknowledged — Vendor has acknowledged the approved Purchase Order.



9. Approval-Level Decision Flow



10. Cohesion and Coupling Justification

10.1 Vendor Registration Module

The Vendor Registration Module should be highly cohesive because its activities are centered on one clear responsibility: managing vendor registration information and the related registration workflow.

10.2 PO Approval Module

The PO Approval Module should also be highly cohesive because its activities focus on Purchase Order creation/submission, approval-level handling, approval decisions, and status updates.

10.3 Coupling Between the Modules

Vendor Registration and PO Approval should remain loosely coupled. The PO process may depend on valid vendor information, but the internal processing of vendor registration should not be tightly embedded inside PO approval logic. This makes the modules easier to understand, test, maintain, and change independently.

11. Design Assumptions and Notes

- The case study's supplied approval limits, stakeholder names, dependencies, and business facts are treated as authoritative.
- The UML diagrams are design representations and do not introduce new case-study facts.
- Technical entities or implementation details not explicitly specified in the case study should be treated as proposed design choices.
- The final Draw.io diagrams should use the same terminology as the BRD, SOW, SRS, and Project Plan.
- The diagrams should be updated if the approved SRS or prototype workflow changes.

12. Diagram Insertion Checklist

☐ Use Case Diagram inserted from Draw.io

☐ Class Diagram inserted from Draw.io

☐ Sequence Diagram — Raise and Approve a Purchase Order inserted

☐ Activity Diagram inserted from Draw.io

☐ Purchase Order State Diagram inserted from Draw.io

☐ Diagram labels match SRS terminology

☐ Actors and approval levels are consistent with the case study

☐ No unsupported case-study figures or facts introduced